Burden and aetiology of diarrhoeal disease in infants and young children in developing countries (the Global Enteric Multicenter Study, GEMS): a prospective, case-control study

Karen L Kotloff, James P Nataro, William C Blackwelder, Dilruba Nasrin, Tamer H Farag, Sandra Panchalingam, Yukun Wu, Samba O Sow, Dipika Sur, Robert F Breiman, Abu S G Faruque, Anita K M Zaidi, Debasish Saha, Pedro L Alonso, Boubou Tamboura, Doh Sanogo, Uma Onwuchekwa, Byomkesh Manna, Thandavarayan Ramamurthy, Suman Kanungo, John B Ochieng, Richard Omore, Joseph O Oundo, Anowar Hossain, Sumon K Das, Shahnawaz Ahmed, Shahida Qureshi, Farheen Quadri, Richard A Adegbola, Martin Antonio, M Jahangir Hossain, Adebayo Akinsola, Inacio Mandomando, Tacilta Nhampossa, Sozinho Acácio, Kousick Biswas, Ciara E O'Reilly, Eric D Mintz, Lynette Y Berkeley, Khitam Muhsen, Halvor Sommerfelt, Roy M Robins-Browne, Myron M Levine

Summary

Background Diarrhoeal diseases cause illness and death among children younger than 5 years in low-income countries. We designed the Global Enteric Multicenter Study (GEMS) to identify the aetiology and population-based burden of paediatric diarrhoeal disease in sub-Saharan Africa and south Asia.

Methods The GEMS is a 3-year, prospective, age-stratified, matched case-control study of moderate-to-severe diarrhoea in children aged 0–59 months residing in censused populations at four sites in Africa and three in Asia. We recruited children with moderate-to-severe diarrhoea seeking care at health centres along with one to three randomly selected matched community control children without diarrhoea. From patients with moderate-to-severe diarrhoea and controls, we obtained clinical and epidemiological data, anthropometric measurements, and a faecal sample to identify enteropathogens at enrolment; one follow-up home visit was made about 60 days later to ascertain vital status, clinical outcome, and interval growth.

Findings We enrolled 9439 children with moderate-to-severe diarrhoea and 13 129 control children without diarrhoea. By analysing adjusted population attributable fractions, most attributable cases of moderate-to-severe diarrhoea were due to four pathogens: rotavirus, *Cryptosporidium*, enterotoxigenic *Escherichia coli* producing heat-stable toxin (ST-ETEC; with or without co-expression of heat-labile enterotoxin), and *Shigella*. Other pathogens were important in selected sites (eg, *Aeromonas*, *Vibrio cholerae* O1, *Campylobacter jejuni*). Odds of dying during follow-up were 8·5-fold higher in patients with moderate-to-severe diarrhoea than in controls (odd ratio 8·5, 95% CI 5·8–12·5, p<0·0001); most deaths (167 [87·9%]) occurred during the first 2 years of life. Pathogens associated with increased risk of case death were ST-ETEC (hazard ratio [HR] 1·9; 0·99–3·5) and typical enteropathogenic *E coli* (HR 2·6; 1·6–4·1) in infants aged 0–11 months, and *Cryptosporidium* (HR 2·3; 1·3–4·3) in toddlers aged 12–23 months.

Interpretation Interventions targeting five pathogens (rotavirus, *Shigella*, ST-ETEC, *Cryptosporidium*, typical enteropathogenic *E coli*) can substantially reduce the burden of moderate-to-severe diarrhoea. New methods and accelerated implementation of existing interventions (rotavirus vaccine and zinc) are needed to prevent disease and improve outcomes.

Funding The Bill & Melinda Gates Foundation.

Published Online
May 14, 2013
http://dx.doi.org/10.1016/S0140-6736(13)60844-2

See Online/Comment
http://dx.doi.org/10.1016/S0140-6736(13)60941-1

Center for Vaccine Development (K L Kotloff MD, J P Nataro MD, W C Blackwelder PhD, D Nasrin PhD, T H Farag PhD, S Panchalingham PhD, Y Wu PhD, L Y Berkeley PhD, K Muhsen PhD, M M Levine MD)**, Department of Pediatrics** (K L Kotloff, J P Nataro, M M Levine)**, and Department of Medicine** (K L Kotloff, J P Nataro, W C Blackwelder, D Nasrin, T H Farag, S Panchalingam, Y Wu, K Muhsen, M M Levine)**, University of Maryland School of Medicine, Baltimore, MD, USA; Department of Pediatrics, University of Virginia School of Medicine, Charlottesville, VA, USA** (J P Nataro)**; Centre pour le Développement des Vaccins, Bamako, Mali** (S O Sow MD, B Tamboura PharmD, D Sanogo MD, U Onwuchekwa MS)**; National Institute of Cholera and Enteric Diseases, Kolkata, India** (D Sur MD, B Manna PhD, T Ramamurthy PhD, S Kanungo MBBS)**; Global Disease Detection Division, Kenya Office of the US Centers for Disease Control and Prevention, Nairobi, Kenya** (R F Breiman MD)**; International Centre for Diarrhoeal Disease Research, Mohakhali, Dhaka, Bangladesh** (A S G Faruque MD, A Hossain MD, S K Das MBBS, S Ahmed MBBS)**; Department of Paediatrics and Child Health, Aga Khan University, Karachi, Pakistan** (A K M Zaidi MBBS, S Qureshi MSc, F Quadri MBBS)**; Medical Research Council (UK) Unit, Fajara, The Gambia** (D Saha MS, R A Adegbola PhD,

Introduction

Globally, one in ten child deaths result from diarrhoeal disease during the first 5 years of life, resulting in about 800 000 fatalities worldwide annually, most occurring in sub-Saharan Africa and south Asia.[1] Although diarrhoeal mortality remains unacceptably high, it is decreasing by about 4% per year,[1] whereas disease incidence is declining more modestly.[2] Interventions that target the main causes and focus on the most susceptible children should further accelerate these declines. To guide these efforts, robust data characterising the burden, risk factors, microbiological aetiology, sequelae, and case fatality of most life-threatening and disabling episodes are essential; heretofore, such data have been scarce in regions with the highest child mortality. To address these knowledge gaps, we created the Global Enteric Multicenter Study (GEMS),[3] the capstone component of which is a 3-year, prospective, age-stratified, matched case-control study of moderate-to-severe diarrhoea in children aged 0–59 months residing in censused populations and seeking care at medical facilities serving seven sites in sub-Saharan Africa and South Asia.[4] We used a common research protocol with standardised epidemiological and microbiological methods to facilitate inter-site comparisons and allow aggregate estimates of aetiology and incidence.[4–6]

M Antonio PhD, M J Hossain MBBS, A Akinsola MD); **Centro de Investigação em Saúde da Manhiça, Maputo, Mozambique** (P L Alonso MD, I Mandomando BS, T Nhampossa MD, S Acácio MD); **Barcelona Centre for International Health Research (CRESIB, Hospital Clínic-Universitat de Barcelona), Barcelona, Spain** (P L Alonso); **Kenya Medical Research Institute/Centers for Disease Control and Prevention (KEMRI/CDC), Kisumu, Kenya** (J B Ochieng MSc, R Omore MCHD, J O Oundo PhD); **GlaxoSmithKline Biologicals, Global Medical Affairs, Wavre, Belgium** (R A Adegbola); **Daffodils Pediatrics and Family Medicine, Tucker, GA, USA** (A Akinsola); **Instituto Nacional de Saúde, Ministério de Saúde, Maputo, Mozambique** (I Mandomando, T Nhampossa, S Acácio); **Department of Veterans Affairs, Cooperative Studies Program Coordinating Center, Perry Point, MD, USA** (K Biswas PhD); **Division of Foodborne, Waterborne, and Environmental Diseases, US Centers for Disease Control and Prevention, Atlanta, GA, USA** (C E O'Reilly PhD, E D Mintz MD); **Center for Drug Evaluation and Research, Office of Antimicrobial Products, Division of Antiinfective Products, Silver Spring, MD, USA** (L Y Berkeley); **Centre for International Health, University of Bergen, Bergen, Norway** (H Sommerfelt PhD); **Division of Infectious Disease Control, Norwegian Institute of Public Health, Oslo, Norway** (H Sommerfelt); **and Department of Microbiology and Immunology, University of Melbourne, Murdoch Children's Research Institute, Royal Children's Hospital, Parkville, VIC, Australia** (R M Robins-Browne)

Correspondence to: Dr Karen L Kotloff, Center for Vaccine Development, University of Maryland School of Medicine, Baltimore, MD 21201, USA **kkotloff@medicine.umaryland.edu**

For the **clinical protocol** see http://medschool.umaryland.edu/GEMS/

Methods

Study design and participants

The primary objective of GEMS was to measure the population-based burden, microbiological aetiology, and adverse clinical effects (eg, growth faltering and death) of moderate-to-severe diarrhoea in developing countries, overall and by age, pathogen, and site.[4] The rationale and underlying epidemiological assumptions,[3,7] and the clinical or epidemiological,[4] microbiological,[5] data management,[8] and analytical[6] methods have been detailed. Salient points are summarised below.

Seven field sites were selected among countries with moderate-to-high under-5 child mortality in Africa (Kenya, Mali, Mozambique, The Gambia) and Asia (Bangladesh, India, Pakistan).[4] Each site provided a censused population, using a demographic surveillance system (DSS) in which fieldworkers visited each household to record births, deaths, and migrations two to four times every year, supplemented by weekly updates of births and deaths in children 0–59 months. GEMS targeted three age strata: infants (0–11 months), toddlers (12–23 months), and children (24–59 months); for each site and age stratum, we used the median population from DSS rounds during the case-control study for analyses. For case enrolment, sites selected sentinel hospitals or health centres (SHCs) where DSS children sought care for diarrhoeal illnesses.[4]

For 36 months between Dec 1, 2007, and March 3, 2011, all children aged 0–59 months belonging to the DSS population at every site who sought care at a SHC were screened for diarrhoea, defined as three or more loose stools within the previous 24 h.[9] A GEMS clinician assessed each child with diarrhoea for eligibility. To be included, the episode had to be new (onset after ≥7 diarrhoea-free days), acute (onset within the previous 7 days), and fulfil at least one of the following criteria for moderate-to-severe diarrhoea: sunken eyes (confirmed by parent or caretaker as more than normal; loss of skin turgor (abdominal skin pinch with slow [≤2 s] or very slow [>2 s] recoil); intravenous hydration administered or prescribed; dysentery (visible blood in loose stools); or admission to hospital with diarrhoea or dysentery.[4] Although all cases meeting the definition of moderate-to-severe diarrhoea were documented, each site restricted enrolment to about the first nine eligible cases per age stratum per fortnight to maintain a manageable work flow throughout the study; we attempted to enrol 600 analysable case-control pairs per age stratum per site in 36 months, which would provide 80% power (two-sided test, 5% significance level) to find a significant difference for a site-stratum-specific comparison of the proportion of cases and controls in whom a specific enteropathogen is identified, if a pathogen is identified in 5·8% of cases and 2·5% of controls.[4,6] For every enrolled case of moderate-to-severe diarrhoea, we aimed to enrol one to three control children without diarrhoea during a home visit, following an algorithm that increased the requisite controls according to the number of patients with moderate-to-severe diarrhoea enrolled in that fortnight. Controls, matched to every individual patient with moderate-to-severe diarrhoea by age (±2 months for patients aged 0–11 months and 12–23 months, and ±4 months for patients aged 24–59 months), sex, and residence (same or nearby village or neighbourhood as the patient with diarrhoea), were randomly selected from the site's DSS database and enrolled within 14 days of the index case.[4] Potential controls who had diarrhoea in the previous 7 days were ineligible.

The clinical protocol was approved by ethics committees at the University of Maryland, Baltimore, MD, USA, and at every field site. Written informed consent was obtained from the parent or primary caretaker of each participant before initiation of study activities.

Procedures

At enrolment, parents or primary caretakers of patients with moderate-to-severe diarrhoea and controls underwent standardised interviews to solicit demographic, epidemiological, and clinical information. GEMS staff, trained in standardised anthropometry, measured the child's length or height three times.[4] Medical management at the SHC and clinical condition upon discharge were documented. Fieldworkers made one follow-up visit to every household of every patient with moderate-to-severe diarrhoea or control child about 60 days after enrolment (targeted range 50–90 days) to assess the child's vital status, capture interim medical events, and repeat anthropometric measurements.

At enrolment, each case and control provided at least 3 g of fresh stool, which within 1 h of passage was placed in cold storage until delivery to the laboratory. Additionally, if antibiotics were to be given to patients before stool was produced, we obtained two rectal swabs for bacterial culture pending passage of the whole stool for the remaining assays.[4] Specimens were placed into transport media immediately (rectal swabs) or within 6 h of passage (whole stool aliquots) for bacterial culture and inoculated onto solid media within 18 h thereafter.[4]

Enteropathogens were identified using uniform methods.[5] Bacterial agents (*Salmonella*, *Shigella*, *Campylobacter*, *Aeromonas*, and *Vibrio* spp) were detected using conventional culture techniques.[5] Three putative *Escherichia coli* colonies from every stool were pooled and analysed by multiplex PCR that detect targets for enterotoxigenic (ETEC), enteroaggregative (EAEC), enteropathogenic (EPEC), and enterohaemorrhagic *E coli* (EHEC).[5] The following gene targets defined each *E coli* pathotype: ETEC (either *eltB* for heat-labile toxin [LT], *estA* for heat-stable toxin [ST], or both), ST-ETEC (either *eltB* and *estA*, or *estA* only), typical EPEC (*bfpA* with or without *eae*), atypical EPEC (*eae* without either *bfpA*, *stx1*, or *stx2*), EAEC (*aatA*, *aaiC*, or both), and EHEC (*eae* with *stx1*, *stx2*, or both, and without *bfpA*). Commercial immunoassays detected rotavirus (ELISA ProSpecT Rotavirus kit, Oxoid, Basingstoke, UK) and adenovirus (ProSpecT Adenovirus Microplate (Oxoid); adenovirus-

positive samples were tested for enteric adenovirus serotypes 40 and 41 (Premier Adenoclone kit, Meridian Bioscience, Cincinnati, OH, USA). Norovirus (genotypes I and II), sapovirus, and astrovirus were detected using multiplex reverse transcriptase (RT) PCR.[5] Individual commercial immunoassays (TechLab, Inc, Blacksburg, VA, USA) detected *Giardia lamblia*, *Entamoeba histolytica* and *Crytosporidium* spp.

Statistical analysis

We assessed associations of moderate-to-severe diarrhoea with potential pathogens using conditional logistic regression[10] with a penalised likelihood approach;[11] taking into account the presence or absence of multiple pathogens as independent variables, we used odds ratios (ORs) and pathogen prevalence among patients with moderate-to-severe diarrhoea to calculate adjusted population attributable fractions (AFs)[12] to estimate pathogen-specific disease burden (expressed as number of cases and incidence rate). The adjusted AF is derived from a multiple conditional logistic regression model that includes other pathogens significantly associated with moderate-to-severe diarrhoea; thus it is the AF adjusted for presence of other pathogens.

Because samples from patients with moderate-to-severe diarrhoea were taken in roughly equal numbers during each fortnight, irrespective of the number of cases of moderate-to-severe diarrhoea appearing at the SHCs, we estimated AF using weights defined as the number of eligible cases divided by the number of enrolled cases—ie, as the inverse of the sampling fraction for cases. We calculated weights separately for cases of moderate-to-severe diarrhoea with and without dysentery, to avoid any bias from overrepresentation or underrepresentation of cases with dysentery. We combined data for two or more adjacent fortnights to avoid having periods with either no patients with diarrhoea and dysentery enrolled or no patients with diarrhoea and without dysentery enrolled; typically, this resulted in the combination of data for eight adjacent periods, so that weighting was based on numbers of cases in 16 weeks.

To estimate disease burden of moderate-to-severe diarrhoea in the population, we did brief surveys of health-care use serially during the case-control study in concert with each round of the DSS, using random samples of children.[11] We asked the parent or primary caretaker of children enrolled in every survey whether her or his child had a new episode of moderate-to-severe diarrhoea during the preceding 7 days, and, if so, the type of health care sought. After pooling data from serial surveys at each site, and applying sampling weights for surveyed children based on the number of children in

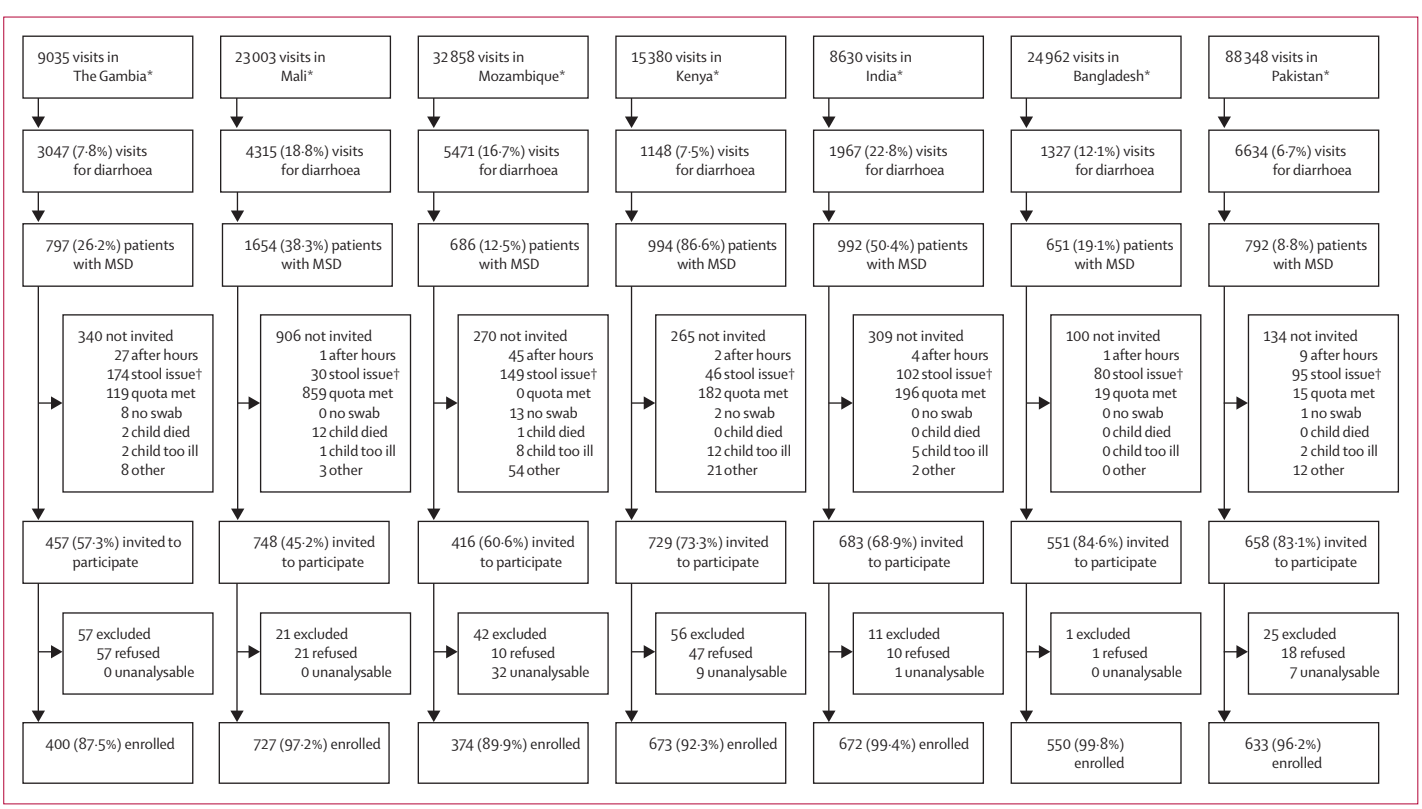

Figure 1: **Study profile of children aged 0–11 months, by site**

*Total visits to a sentinel health center by children in the demographic surveillance system area belonging to the corresponding age stratum. †Stool issues include no specimen, insufficient specimen, and improperly handled specimen.

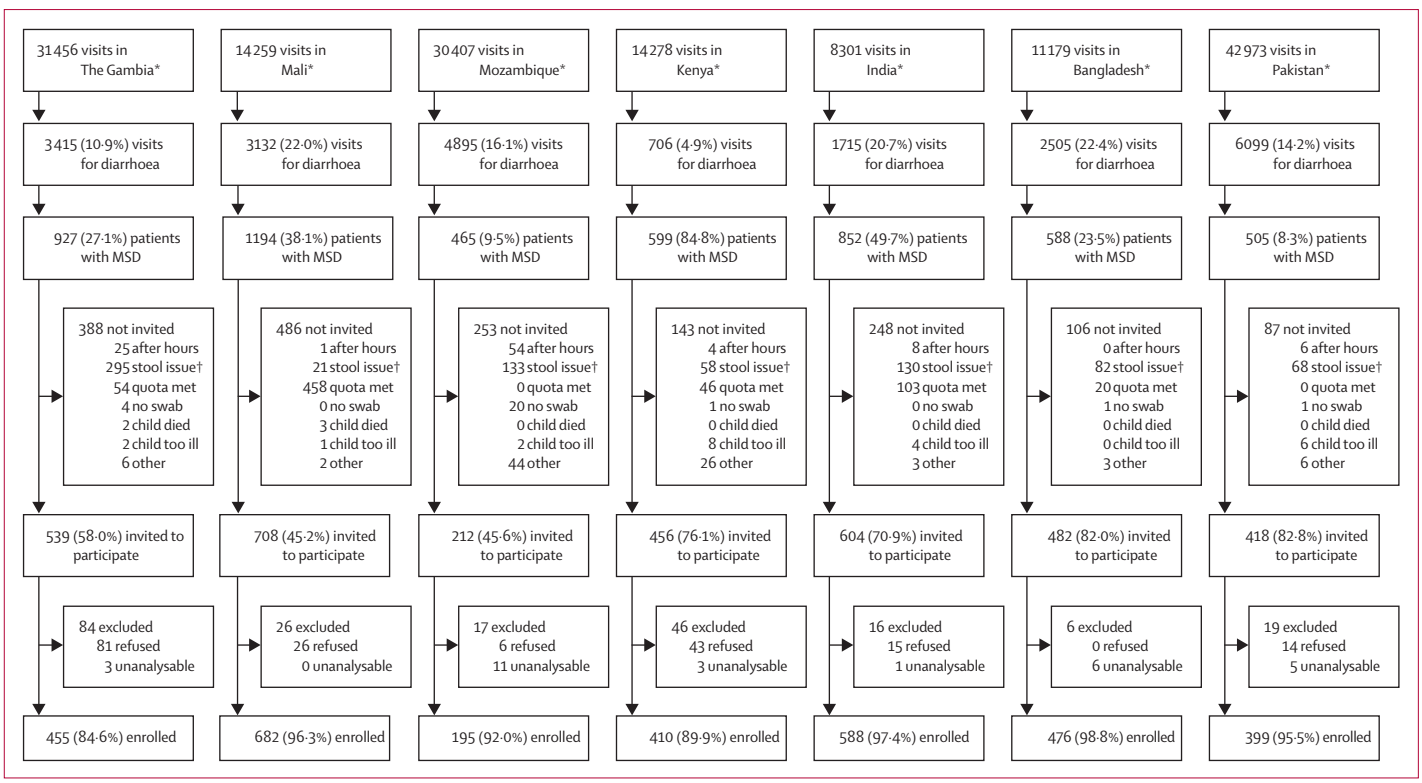

Figure 2: **Study profile of children aged 12–23 months, by site**

*Total visits to a sentinel health center by children in the demographic surveillance system area belonging to the corresponding age stratum. †Stool issues include no specimen, insufficient specimen, and improperly handled specimen.

each age–sex stratum in the DSS population, we calculated the proportion, designated "r", of children with moderate-to-severe diarrhoea who were taken to an SHC at each site within 7 days of onset of diarrhoea.

For each site and age stratum, we estimated numbers of cases and incidence rates of moderate-to-severe diarrhoea per 100 child-years as follows. We calculated the annual number of cases of moderate-to-severe diarrhoea in the population as the number of eligible cases of moderate-to-severe diarrhoea recorded at SHCs during the 36-month study, divided by 3×r (with "r" defined as above). Division of this result by the median DSS population gave the moderate-to-severe diarrhoea incidence rate. To derive the number of cases and the incidence of moderate-to-severe diarrhoea attributable to a specific pathogen, the total cases and incidence rates of moderate-to-severe diarrhoea were multiplied by the pathogen's weighted AF. Additionally, we calculated the incidence rates for all moderate-to-severe diarrhoea and for pathogen-specific attributable moderate-to-severe diarrhoea across sites by adding all cases of moderate-to-severe diarrhoea or all pathogen-specific attributable cases across sites and dividing by the sum of the sites' populations.

See Online for appendix

We compared proportions of cases and controls who died during follow-up using conditional logistic regression or, when numbers of deaths were small (India and Bangladesh), Fisher's exact test. We calculated associations of pathogens with risk of dying in patients with moderate-to-severe diarrhoea, both unadjusted and adjusted for other pathogens and for site, using weighted proportional hazards (Cox) regression models to allow for different durations of follow-up, using the same weights as for analysis of AF.

The primary measure of growth in our analyses was the length or height-for-age *Z* (HAZ) score,[13] which we derived using the median of three repeated measurements for every child at every visit according to WHO standards.[14] We deleted implausible values for height and values that were inconsistent between enrolment and follow-up (appendix). Our analyses included only case-control sets with data on both enrolment and follow-up HAZ for the patients with moderate-to-severe diarrhoea. We calculated weighted means of enrolment HAZ score and change in HAZ score from enrolment to follow-up (δHAZ) for patients with moderate-to-severe diarrhoea and controls, using the same type of weights as for AF analysis, redefined for the HAZ dataset. We compared enrolment HAZ scores for both controls and patients with moderate-to-severe diarrhoea using weighted paired *t* tests; we compared δHAZ using weighted linear regression models for all possible matched pairs, adjusting for enrolment HAZ score and duration of follow-up, using jackknife estimates of standard error.

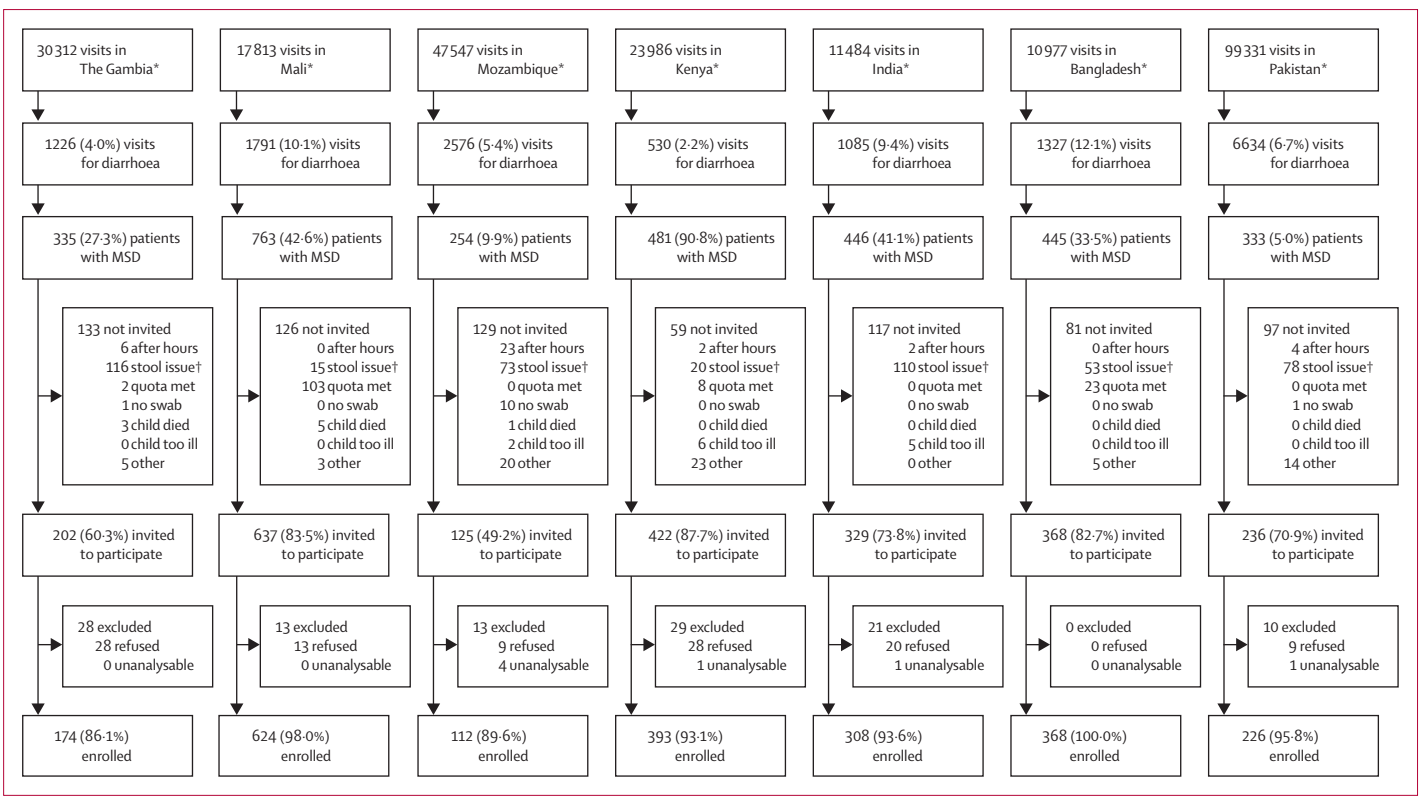

Figure 3: **Study profile of children aged 24–59 months, by site**
*Total visits to a sentinel health center by children in the demographic surveillance system area belonging to the corresponding age stratum. †Stool issues include no specimen, insufficient specimen, and improperly handled specimen.

We made no imputation of missing values. In regression models, we excluded observations with missing values from analysis. We deemed two-sided p values of 0·05 or lower to be significant.

Role of the funding source

The sponsor of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

During the 36-month case-control study at seven DSS sites, children aged 0–59 months cumulatively contributed about 487 386 child-years of observation. They made 626 519 visits to an SHC, of which 66 009 (11%) were by children with diarrhoea (12·2% of the visits for infants, 14·7% of the visits for toddlers, and 6·3% of the visits for children); 14 753 children met enrolment criteria for moderate-to-severe diarrhoea (22% of diarrhoea cases), of whom 9980 were invited to participate (68% of eligible; figures 1–3). Of those invited, 9439 were included in this analysis (95%); 456 refused to participate (5%) and 85 did not have a matched control (1%) and were deemed unanalysable. Concomitantly, we enrolled 13 129 matched controls. The household was visited about 60 days after enrolment for 8549 (91%) of patients with moderate-to-severe diarrhoea known to be alive at discharge from the SHC and 12 390 (94%) of enrolled control children. When we compared demographic and health indicators in cases and controls (table 1), controls in several sites and age groups belonged to a higher wealth quintile (eg, India and Pakistan), had greater access to improved water (Mali, Kenya, and The Gambia) and had more educated mothers (India) than did patients with moderate-to-severe diarrhoea.

Overall, we identified one or more putative pathogens in 7851 (83%) children with moderate-to-severe diarrhoea and in 9395 (72%) controls; two or more agents were identified in 4200 (45%) cases and 4075 (31%) controls. If one considers only the pathogens significantly associated with moderate-to-severe diarrhoea by conditional logistic regression and calculates the AF for the group of pathogens as a whole,[6] the median proportion of episodes attributable to a pathogen was 44% (IQR 41–52) for infants, 47% (21–52) for toddlers, and 40% (23–53) for children. For some pathogens (eg, rotavirus, *Shigella*, *V cholerae* O1, adenovirus serotypes 40/41), nearly all infected children were symptomatic with moderate-to-severe diarrhoea, so a high percentage (about 90%) of cases of moderate-to-severe diarrhoea with the pathogen were attributable to that pathogen, compared with

	Basse, The Gambia		Bamako, Mali		Manhiça, Mozambique		Nyanza Province, Kenya		Kolkata, India		Mirzapur, Bangladesh		Karachi (Bin Qasim Town), Pakistan	
	Cases	Controls	Cases	Controls	Cases	Controls	Cases	Controls	Cases	Controls	Cases	Controls	Cases	Controls
0–11 months														
Number of patients	400	585	727	727	374	697	673	673	672	685	550	878	633	633
Mean age in months (SD)	7·8 (2·4)	7·9 (2·2)	7·3 (2·7)	7·0* (2·5)	6·5 (2·9)	6·7 (2·8)	6·5 (2·7)	6·3* (2·5)	6·8 (2·9)	6·8 (2·8)	7·2 (2·6)	7·0* (2·6)	6·1 (2·9)	6·0* (2·8)
Female individuals†	165 (41%)	255 (44%)	328 (45%)	328 (45%)	148 (40%)	272 (39%)	274 (41%)	274 (41%)	299 (45%)	307 (45%)	216 (39%)	346 (39%)	292 (46%)	292 (46%)
Caretaker completed primary school	25 (6%)	29 (5%)	118 (16%)	113 (16%)	84 (23%)	191 (28%)	361 (54%)	335 (50%)	404 (60%)	467 (68%)*	420 (76%)	659 (75%)	106 (17%)	121 (19%)
Mean wealth quintile[15]	1·9	1·9	2·0	2·0	1·9	2·0	2·0	1·8	1·9	2·1*	2·1	2·0*	1·8	2·0*
Number in household, median (range)	24 (3–129)	27* (3–120)	14 (2–71)	15 (3–110)	6 (2–42)	7* (2–73)	5 (2–17)	5 (2–40)	6 (2–18)	6 (3–20)	5 (2–23)	5 (2–30)	8 (2–50)	8 (2–66)
Access to improved water‡	164 (41%)	268 (46%)	471 (65%)	500 (69%)	108 (29%)	215 (31%)	68 (10%)	71 (11%)	328 (49%)	236 (34%)*	543 (99%)	869 (99%)	7 (1%)	4 (1%)
12–23 months														
Number of patients	455	639	682	695	195	391	410	621	588	598	476	761	399	676
Mean age in months (SD)	16·9 (3·4)	17·3* (3·3)	16·8 (3·5)	16·3* (3·2)	16·2 (3·2)	16·7* (3·1)	16·5 (3·5)	16·3 (3·1)	16·6 (3·4)	16·8 (3·1)	16·7 (3·3)	16·7 (3·1)	16·4 (3·3)	16·5 (2·9)
Female individuals†	208 (46%)	284 (44%)	300 (44%)	305 (44%)	85 (44%)	155 (40%)	186 (45%)	279 (45%)	259 (44%)	264 (44%)	209 (44%)	341 (45%)	166 (42%)	282 (42%)
Caretaker completed primary school	27 (6%)	33 (5%)	96 (14%)	103 (15%)	37 (19%)	87 (22%)	223 (54%)	320 (52%)	323 (55%)	379 (63%)*	356 (75%)	579 (76%)	58 (15%)	120 (18%)
Mean wealth quintile[15]	2·0	2·1	2·1	2·0	1·9	1·9	1·9	1·9	1·8	2·0*	1·9	2·0	1·9	2·2*
Number in household, median (range)	23 (3–229)	30* (3–147)	13 (2–150)	13 (3–77)	6 (2–39)	6 (2–24)	5 (2–13)	6* (2–16)	6 (3–25)	5 (3–25)	5 (2–18)	5 (2–20)	8 (2–43)	8 (2–36)
Access to improved water	205 (45%)	289 (45%)	442 (65%)	492 (71%)*	57 (29%)	103 (27%)	33 (8%)	75 (12%)*	272 (46%)	208 (35%)*	472 (99%)	754 (99%)	2 (1%)	2 (0%)
24–59 months														
Number of patients	174	345	624	642	112	208	393	589	308	731	368	826	226	529
Mean age in months (SD)	31·8 (8·4)	31·6* (7·3)	35·6 (9·7)	35·1* (9·5)	33·8 (9·2)	33* (8·3)	36·1 (9·5)	35·8 (9·2)	35·6 (9·8)	35·6 (9·4)	34·9 (8·6)	34·9 (8·6)	35·0 (9·7)	35·4* (9·6)
Female individuals†	86 (49%)	174 (50%)	282 (45%)	290 (45%)	48 (43%)	74 (36%)	177 (45%)	266 (45%)	123 (40%)	299 (41%)	155 (42%)	355 (43%)	89 (39%)	209 (40%)
Caretaker completed primary school	16 (9%)	16 (5%)*	92 (15%)	92 (14%)	24 (22%)	45 (22%)	219 (56%)	320 (54%)	186 (60%)	458 (63%)	271 (74%)	608 (74%)	43 (19%)	103 (20%)
Mean wealth quintile[15]	2·1	2·0	2·1	1·9*	2·0	2·2	2·1	2·1	1·8	2·1*	2·0	2·1	1·7	2·2*
Number in household, median (range)	23 (3–100)	30* (3–180)	15 (2–100)	13* (2–81)	6 (2–16)	6* (3–72)	5 (2–14)	6* (2–25)	5 (3–15)	5 (2–18)	5 (2–18)	5* (2–22)	8 (3–26)	8 (2–60)
Access to improved water	62 (36%)	155 (45%)*	414 (66%)	446 (69%)	37 (33%)	58 (28%)	39 (10%)	60 (10%)	150 (49%)	248 (34%)*	366 (99%)	822 (100%)	0	0

SD=Standard deviation. *Significant difference, by conditional logistic regression (p≤0·05). †Statistical comparison not done because of exact sex matching; differences in overall proportions are due to numbers of multiple controls that differ by sex. ‡Improved water: the main source of drinking water for the household is either piped (into house or yard), public tap, tubewell (deep or shallow), covered well, protected spring, rainwater, or borehole, and is accessible within 15 min or less, roundtrip, and is available daily.

Table 1: **Demographic features of cases with moderate-to-severe diarrhoea and their matched control**

60–70% for agents such as *Cryptosporidium*, ST-ETEC, and *Aeromonas*.

Four pathogens were significantly associated with moderate-to-severe diarrhoea at all seven study sites in one or more age strata: rotavirus, *Cryptosporidium*, *Shigella*, and ST-ETEC (ST-only or LT/ST strains) (table 2). Most attributable episodes were associated with one of these pathogens. Rotavirus had the highest AF of any pathogen at every site during infancy. Although its AF generally diminished with age, rotavirus had the largest AF of any pathogen in toddlers at four sites, and at the Mali and India sites even in the eldest stratum. *Cryptosporidium* had the second highest AF during infancy at five sites, persisting in importance, albeit at a lower level, during the second year of life at five sites; in the eldest stratum, *Cryptosporidium* was significantly associated with diarrhoea only in Kenya. By contrast, the adjusted AF of *Shigella* increased from infants to toddlers at every site, rising to the rank of first or second in AF at four sites in toddlers and five sites in the eldest stratum. In Mirzapur, Bangladesh, a GEMS site with rather high maternal education, low household crowding, and the

	Basse, The Gambia	Bamako, Mali	Manhiça, Mozambique	Nyanza Province, Kenya	Kolkata, India	Mirzapur, Bangladesh	Karachi (Bin Qasim Town), Pakistan
0–11 months							
Number of patients	400	727	374	673	672	550	633*
Rotavirus	23·5 (18·8–28·2)	21·7 (18·3–25·2)	27·8 (21·0–34·6)	19·7 (16·4–23·1)	27·0 (23·3–30·6)	16·3 (12·8–19·8)	22·6 (18·9–26·3)
Cryptosporidium	11·7 (7·6–15·7)	14·0 (10·5–17·6)	14·7 (9·6–19·9)	9·0 (5·7–12·3)	11·8 (8·0–15·6)	5·3 (2·1–8·5)	5·6 (1·4–9·8)
ST-ETEC (ST-only or LT/ST)	4·9 (1·2–8·7)	3·6 (1·7–5·5)	··	7·0 (3·9–10·0)	3·0 (1·3–4·7)	1·4 (0·2–2·6)	7·0 (4·3–9·7)
Shigella	4·0 (1·7–6·4)	··	··	4·5 (2·4–6·6)	2·0 (0·7–3·3)	13·2 (10·3–16·1)	7·6 (5·3–9·9)
Norovirus GII	8·9 (4·3–13·4)	··	··	··	··	··	··
Aeromonas	··	··	··	··	··	9·7 (4·1–15·3)	11·3 (6·9–15·7)
Adenovirus 40/41	2·3 (0·6–4·0)	1·9 (0·8–3·0)	2·2 (0·4–4·0)	··	4·0 (2·3–5·6)	3·9 (2·0–5·7)	1·8 (0·8–2·9)
Campylobacter jejuni	··	··	··	-	··	9·0 (1·7–16·4)	6·7 (1·0–12·4)
Typical EPEC	··	··	··	5·2 (1·8–8·5)	··	··	··
Non-typhoidal *Salmonella*	··	··	··	··	··	4·2 (2·2–6·2)	··
Vibrio cholerae O1	··	··	··	··	··	··	3·1 (1·5–4·7)
Entamoeba histolytica	··	··	··	··	··	3·4 (0·4–6·4)	··
AF for all associated pathogens‡	51·9 (44·9–58·8)	39·5 (34·7–44·2)	41·6 (34·0–49·2)	41·1 (35·6–46·5)	43·7 (39·0–48·5)	58·4 (52·6–64·2)	52·3 (46·2–58·5)
12–23 months							
Number of patients	455	682	195	410	588	476	399
Rotavirus	17·0 (13·4–20·6)	11·8 (9·0–14·7)	··	13·3 (9·7–17·0)	25·4 (21·5–29·2)	18·3 (14·5–22·1)	9·8 (6·4–13·3)
Shigella	12·8 (8·7–16·9)	2·4 (0·8–4·0)	6·6 (2·5–10·7)†	4·6 (1·6–7·6)	7·2 (4·7–9·7)	52·2 (47·5–56·8)	12·5 (8·5–16·5)
Cryptosporidium	7·7 (4·5–10·9)	4·7 (1·0–8·3)	··	8·9 (5·4–12·4)	8·4 (3·6–13·2)	··	8·2 (4·3–12·2)
ST-ETEC (ST-only or LT/ST)	8·0 (3·1–13·0)	2·3 (0·2–4·4)	9·0 (3·6–14·3)	6·9 (3·6–10·3)	5·8 (3·1–8·5)	··	5·7 (2·4–8·9)
Aeromonas	··	··	··	··	··	11·9 (3·9–19·9)	9·5 (3·2–15·8)
Norovirus GII	8·7 (5·2–12·1)	··	··	··	4·7 (1·3–8·1)	··	··
Vibrio cholerae O1	··	··	··	··	3·4 (1·7–5·0)	1·4 (0·3–2·5)	7·5 (4·8–10·2)
Adenovirus 40/41	2·2 (0·7–3·7)	··	··	··	4·5 (2·7–6·3)	··	2·2 (0·2–4·1)
Non-typhoidal *Salmonella*	··	··	··	3·2 (0·5–6·0)	··	··	··
Typical EPEC	··	··	··	3·5 (0·3–6·7)	··	··	··
EAEC	··	··	··	··	··	9·9 (2·0–17·8)	··
AF for all associated pathogens‡	49·3 (43·0–55·6)	20·7 (15·8–25·7)	15·5 (8·9–22·2)	36·8 (30·2–43·4)	52·2 (46·5–57·8)	75·9 (70·7–81·2)	47·1 (40·2–54·1)
24–59 months							
Number of patients	174	624	112	393	308	368	226
Shigella	12·6 (6·4–18·8)	2·0 (0·0–19·5)	14·9 (7·9–21·9)	9·6 (6·1–13·1)	12·1 (7·8–16·4)	67·6 (61·3–73·8)	10·0 (4·6–15·4)
Aeromonas	··	··	··	··	··	18·3 (10·8–25·9)	24·1 (16·0–32·3)
Rotavirus	12·1 (6·6–17·6)	3·0 (0·0–21·9)	–	3·5 (1·3–5·6)	14·5 (10·2–18·7)	··	··
Vibrio cholerae O1	··	··	8·3 (2·8–13·9)	··	7·6 (4·6–10·6)	3·0 (1·0–5·0)	12·1 (7·7–16·5)
ST-ETEC (ST-only or LT/ST)	9·2 (3·1–15·3)	··	··	4·9 (2·0–7·9)	6·1 (3·1–9·1)	··	5·8 (1·2–10·3)
Campylobacter jejuni	··	··	··	··	9·9 (4·9–14·9)	··	16·1 (6·5–25·7)
Entamoeba histolytica	··	2·0 (0·0–18·5)	··	··	··	··	··
Norovirus GII	9·4 (2·6–16·2)	··	··	··	··	··	··
Non-typhoidal *Salmonella*	··	··	··	3·7 (1·2–6·1)	··	··	··
Cryptosporidium	··	··	··	2·5 (0·2–4·9)	··	··	··
Sapovirus	··	··	··	··	3·5 (1·2–5·8)	··	··
AF for all associated pathogens‡	39·9 (30·2–49·6)	6·8 (0·0–29·3)	23·3 (14·9–31·7)	23·3 (17·8–28·8)	46·9 (39·9–53·8)	75·6 (69·3–81·8)	52·6 (43·3–61·9)

ST=heat stable toxin. LT=heat labile toxin. ETEC=enterotoxigenic *Escherichia coli*. EPEC=enteropathogenic *E coli*. AF=adjusted attributable fraction. MSD=moderate-to-severe diarrhoea. Pathogens included in the table are those that were significantly associated with MSD in weighted multiple conditional logistic regression analysis. *Astrovirus, though not included in table 2, was marginally significantly associated with MSD among infants in Pakistan (OR=1·8, p=0·0501). †Included in conditional logistic regression model even though not significant by jackknife because of a small number of controls (*Shigella* was isolated in 12/195 cases and 1/391 controls; p<0·0001 by Fisher's exact test). ‡The total attributable MSD for each stratum was determined from a model with all associated pathogens included;[22] it is not the sum of individual pathogen-specific AFs.

Table 2: **Adjusted attributable fraction (AF, expressed as weighted percent of total episodes with 95% CI) of pathogens significantly associated with moderate-to-severe diarrhoea (MSD), by age stratum and site**

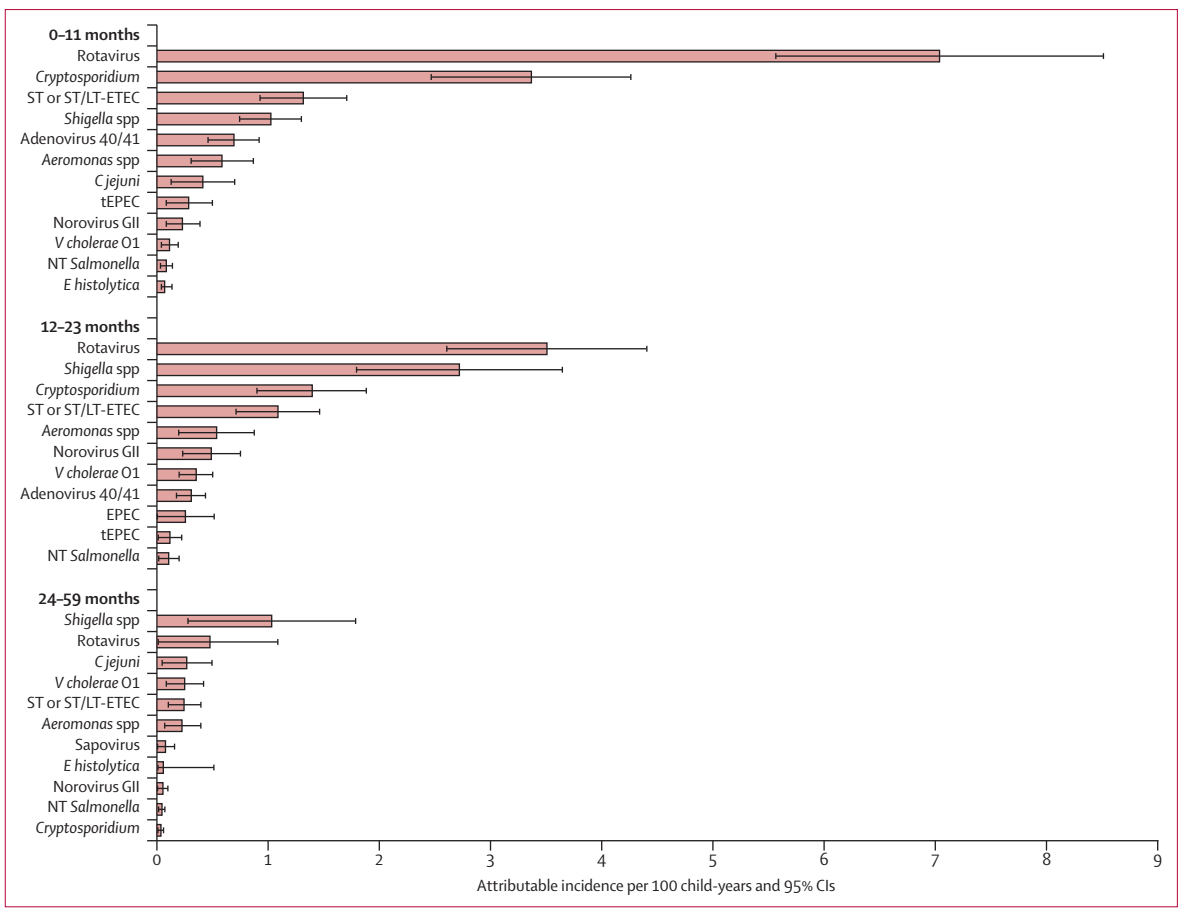

Figure 4: **Attributable incidence of pathogen-specific moderate-to-severe diarrhoea per 100 child-years by age stratum, all sites combined**
The bars show the incidence rates and the error bars show the 95% CIs.

highest reported access to improved water (table 1), *Shigella* was frequently isolated in patients with moderate-to-severe diarrhoea with dysentery (16·6% for infants, 66·0% for toddlers, and 78·4% for children) and in cases with watery diarrhoea (2·9% for infants, 20·6% for toddlers, and 43·4% for children). ST-ETEC was a significant pathogen at every site in at least one age stratum and in all age strata at four sites. By contrast, ETEC producing LT alone was not a significant cause of moderate-to-severe diarrhoea at any site or age stratum. A small proportion (<5%) of moderate-to-severe diarrhoea was attributable to adenovirus 40/41 at six sites during infancy, and in three sites during the second year of life.

Three enteropathogens showed regional importance. *Aeromonas* was a leading pathogen in the Pakistan and Bangladesh sites, with the peak AF at age 24–59 months. *V cholerae* O1 appeared in an age-escalating pattern in the three Asian sites plus Mozambique. *C jejuni* was significantly associated with moderate-to-severe diarrhoea in at least one age stratum at the three Asian sites.

Several pathogens had a smaller distribution, and were significantly associated with moderate-to-severe diarrhoea in two or fewer sites per age stratum, including norovirus (GII genogroup), sapovirus, EAEC, typical EPEC, non-typhoidal *Salmonella*, and *E histolytica* (table 2). *Giardia* was not significantly positively associated with moderate-to-severe diarrhoea; to the contrary, in univariate analyses *Giardia* was identified significantly more frequently in controls than in patients with moderate-to-severe diarrhoea aged 12–59 months in ten of the 14 age-site strata.

When we estimated DSS-wide annual incidence rates of moderate-to-severe diarrhoea at all sites combined, rotavirus dominated during the first 2 years of life, with an incidence of moderate-to-severe diarrhoea during infancy (7·0 episodes per 100 child-years, 95% CI 5·4–8·5) that was more than double that of any other pathogen (figure 4). Generally, we noted two pathogens per age stratum whose incidence markedly exceeded the others: rotavirus and *Cryptosporidium* in infants; rotavirus and *Shigella* in toddlers; and *Shigella* and rotavirus in children (figure 4).

Regardless of the age stratum, the estimated incidence of moderate-to-severe diarrhoea was highest in India, next highest in Kenya and Mali, and lowest in The Gambia, Pakistan, Bangladesh, and Mozambique (table 3). The overall annual incidence of moderate-to-severe diarrhoea per 100 child-years was 30·8 (95% CI

24·8–36·8) for infants, 23·1 (95% CI 17·2–29·0) for toddlers, and 7·7 (95% CI 3·9–11·5) for children.

During follow-up within 90 days of enrolment, 190 (2·0%) deaths were detected in the 9439 children enrolled with moderate-to-severe diarrhoea, and 37 (0·3%) deaths were detected in the 13 129 control children (OR 8·5, 95% CI 5·8–12·5, p<0·0001; table 4). Mortality in children with moderate-to-severe diarrhoea was highest in the Mozambique site, followed by The Gambia and Kenya, Pakistan and Mali, and finally Bangladesh and India (table 4). Mortality in patients with moderate-to-severe diarrhoea exceeded mortality in controls at all sites and the differences were significant everywhere except in India (table 4). In patients with moderate-to-severe diarrhoea, 64 (34%) of deaths occurred on days 0–7 after enrolment, 63 (33%) on days 8–21, and 63 (33%) after day 21; controls survived significantly longer than did patients with moderate-to-severe diarrhoea (p<0·0001 by logrank test). Although 49 (26%) of deaths in patients with moderate-to-severe diarrhoea occurred during the enrolment encounter at the SHC and 36 (19%) during a subsequent medical contact, importantly, 105 (55%) occurred at home or outside of a medical facility.

Most deaths in patients with moderate-to-severe diarrhoea occurred in infants (107 [56%]) and toddlers (60 [32%]). Even so, the weighted risk of mortality remained high in the oldest stratum in The Gambia (1·8%), Kenya (2·3%), and Mozambique (3·9%). In multiple Cox regression analysis, pathogens associated

	Basse, The Gambia	Bamako, Mali	Manhiça, Mozambique	Nyanza Province, Kenya	Kolkata, India	Mirzapur, Bangladesh	Karachi (Bin Qasim Town), Pakistan
0–11 months							
MSD–total*	13·4 (7·8–19·1)	38·7 (17·0–60·5)	12·5 (6·2–18·8)	51·3 (29·0–73·6)	94·3 (56·4–132·2)	12·7 (6·5–18·9)	24·5 (12·2–36·8)
MSD–attributable	7·0 (3·9–10·0)	15·3 (6·5–24·1)	5·2 (2·4–8·0)	21·1 (11·5–30·6)	41·2 (24·1–58·4)	7·4 (3·7–11·1)	12·8 (6·2–19·4)
1	Rotavirus 3·2 (1·7–4·6)	Rotavirus 8·4 (3·5–13·3)	Rotavirus 3·5 (1·5–5·4)	Rotavirus 10·1 (5·4–14·8)	Rotavirus 25·4 (14·7–36·2)	Rotavirus 2·1 (1·0–3·2)	Rotavirus 5·5 (2·6–8·5)
2	*Cryptosporidium* 1·6 (0·7–2·4)	*Cryptosporidium* 5·4 (2·1–8·8)	*Cryptosporidium* 1·8 (0·7–3·0)	*Cryptosporidium* 4·6 (2·0–7·2)	*Cryptosporidium* 11·1 (5·4–16·9)	*Shigella* 1·7 (0·8–2·6)	*Aeromonas* 2·8 (1·0–4·5)
3	Norovirus GII 1·2 (0·4–2·0)	ST-ETEC 1·4 (0·3–2·5)	Adenovirus 40/41 0·3 (0·0–0·5)	Typical EPEC 2·7 (0·6–4·7)	Adenovirus 40/41 3·7 (1·6–5·9)	*C jejuni* 1·1 (0·1–2·2)	*Shigella* 1·9 (0·8–2·9)
4	ST-ETEC 0·7 (0·1–1·2)	Adenovirus 40/41 0·7 (0·1–1·3)	..	ST-ETEC 3·6 (1·4–5·8)	ST-ETEC 2·8 (0·9–4·8)	Cryptosporidium 0·7 (0·2–1·2)	ST-ETEC 1·7 (0·6–2·8)
5	*Shigella* 0·5 (0·2–0·9)	..	..	*Shigella* 2·3 (0·8–3·8)	*Shigella* 1·9 (0·4–3·3)	Adenovirus 40/41 0·5 (0·2–0·8)	*C jejuni* 1·7(0·0–3·3)
6	Adenovirus 40/41 0·3 (0·1–0·6)	..	..	..	..	NT *Salmonella* 0·5 (0·2–0·9)	*Cryptosporidium* 1·4 (0·1–2·6)
7	..	..	..	..	..	*E histolytica* 0·5 (0·0–0·9)	Adenovirus 40/41 0·5 (0·1–0·8)
8	..	..	..	..	..	ST-ETEC 0·2 (0·0–0·4)	*V cholerae* O1 0·8 (0·2–1·3)
9	..	..	..	..	..	Aeromonas 1·2 (0·3–2·2)	..
12–23 months							
MSD–total*	19·2 (8·4–30·0)	34·3 (10·0–58·7)	7·8 (4·9–10·7)	22·1 (14·1–30·2)	48·8 (29·3–68·2)	16·2 (6·3–26·1)	16·6 (7·3–25·9)
MSD–attributable	9·5 (4·0–14·9)	7·1(1·8–12·5)	1·2 (0·5–1·9)	8·2 (4·8–11·5)	25·4 (14·9–36·0)	12·3 (4·8–19·9)	7·8 (3·3–12·3)
1	Rotavirus 3·3 (1·3–5·2)	Rotavirus 4·1 (1·0–7·1)	ST-ETEC 0·7 (0·2–1·2)	Rotavirus 3·0 (1·6–4·3)	Rotavirus 12·4 (7·1–17·7)	*Shigella* 8·5 (3·3–13·7)	*Shigella* 2·1 (0·7–3·4)
2	*Shigella* 2·5 (0·9–4·1)	*Cryptosporidium* 1·6 (0·0–3·3)	*Shigella* 0·5 (0·1–0·9)	*Cryptosporidium* 2·0 (0·9–3·0)	*Shigella* 3·5 (1·7–5·4)	Rotavirus 3·0 (1·1–4·9)	*Aeromonas* 1·6 (0·2–2·9)
3	Norovirus GII 1·7 (0·5–2·8)	ST-ETEC 0·8 (0·0–1·7)	..	ST-ETEC 1·5 (0·6–2·5)	ST-ETEC 2·8 (1·1–4·6)	*Aeromonas* 1·9 (0·2–3·7)	Rotavirus 1·6 (0·6–2·7)
4	*Cryptosporidium* 1·5 (0·4–2·5)	*Shigella* 0·8 (0·0–1·6)	..	*Shigella* 1·0 (0·3–1·8)	Norovirus GII 2·3 (0·4–4·2)	*V cholerae* O1 0·2 (0·0–0·5)	*Cryptosporidium* 1·4 (0·4–2·4)
5	ST-ETEC 1·5 (0·3–2·8)	..	..	Typical EPEC 0·8 (0·0–1·5)	Adenovirus 40/41 2·2 (0·9–3·4)	EAEC 1·6 (0·0–3·2)	*V cholerae* O1 1·3 (0·4–2·1)
6	Adenovirus 40/41 0·4 (0·0–0·8)	..	..	NT *Salmonella* 0·7 (0·1–1·4)	*V cholerae* O1 1·6 (0·6–2·7)	..	ST-ETEC 0·9 (0·2–1·7)
7	..	..	..	..	*Cryptosporidium* 4·1 (1·2–6·9)	..	Adenovirus 40/41 0·4 (0·0–0·7)

(Continues on next page)

	Basse, The Gambia	Bamako,Mali	Manhiça, Mozambique	Nyanza Province, Kenya	Kolkata,India	Mirzapur, Bangladesh	Karachi (Bin Qasim Town), Pakistan
(Continued from previous page)							
24–59 months							
MSD—total*	2·9 (1·0–4·9)	14·6 (0·0–31·3)	2·8 (0·1–5·6)	7·5 (5·3–9·6)	24·1 (0·6–47·6)	4·6 (0·0–9·3)	2·2 (0·8–3·6)
MSD—attributable	1·2 (0·3–2·0)	1·0 (0·0–4·5)	0·7 (0·0–1·3)	1·7 (1·1–2·4)	11·3 (0·1–22·4)	3·4 (0·0–7·1)	1·2 (0·4–1·9)
1	*Shigella* 0·4 (0·1–0·7)	Rotavirus 0·4 (0·0–3·2)	*Shigella* 0·4 (0·0–0·9)	*Shigella* 0·7 (0·4–1·1)	Rotavirus 3·5 (0·0–7·1)	*Shigella* 3·1 (0·0–6·3)	*Aeromonas* 0·5 (0·2–0·9)
2	Rotavirus 0·4 (0·1–0·6)	*Shigella* 0·3 (0·0–2·9)	*V cholerae* O1 0·2 (0·0–0·5)	ST-ETEC 0·4 (0·1–0·6)	*Shigella* 2·9 (0·0–5·9)	*Aeromonas* 0·8 (0·0–1·8)	*C jejuni* 0·4 (0·0–0·7)
3	Norovirus GII 0·3 (0·0–0·5)	*E histolytica* 0·3 (0·0–2·7)	··	Rotavirus 0·3 (0·1–0·4)	*C jejuni* 2·4 (0·0–5·0)	*V cholerae* O1 0·1 (0·0–0·3)	*V· cholerae* O1 0·3 (0·1–0·5)
4	ST-ETEC 0·3 (0·0–0·5)	··	··	NT *Salmonella* 0·3 (0·1–0·5)	*V cholerae* O1 1·8 (0·0–3·8)	··	*Shigella* 0·2 (0·0–0·4)
5	··	··	··	*Cryptosporidium* 0·2 (0·0–0·4)	ST-ETEC 1·5 (0·0–3·1)	··	ST-ETEC 0·1 (0·0–0·3)
6					Sapovirus 0·8 (0·0–1·8)		

MSD=moderate-to-severe diarrhoea; EPEC=enteropathogenic *Escherichia coli*. ST=heat stable toxin. ST-ETEC=either ST or ST/LT producing enterotoxigenic *E coli*. NT=non-typhoidal. AF=adjusted attributable fraction. Pathogens included in the table are those that were significantly associated with MSD in weighted multiple conditional logistic regression analysis. *The total attributable MSD for each stratum was identified for the group of pathogens together[12] and might be exceeded by the sum of each individual pathogen-specific AF, which does not account for co-infections.

Table 3: **Weighted annual incidence (per 100 child-years) of total MSD, MSD attributable to a pathogen, and MSD attributable to a specific pathogen, with 95% confidence interval, by age stratum and site, in ordinal rank for specific pathogens**

	Cases		Controls		OR (95% CI)
	Total*	Number of deaths (%)	Total*	Number of deaths (%)	
Manhiça, Mozambique	681	51 (7·5%)	1296	11 (0·85%)	13·4 (6·1–29·3)†
Basse, The Gambia	1029	39 (3·8%)	1569	7 (0·58%)	7·0 (3·0–16·5)†
Nyanza Province, Kenya	1476	52 (3·5%)	1883	11 (0·50%)	5·5 (2·8–10·7)†
Karachi (Bin Qasim Town), Pakistan	1258	16 (1·3%)	1838	1 (0·05%)	13·1 (0·99–172·4)†
Bamako, Mali	2033	23 (1·1%)	2064	5 (0·24%)	5·5 (1·8–16·5)†
Mirzapur, Bangladesh	1394	7 (0·50%)	2465	1 (0·04%)	12·4 (2·0–77·5)‡
Kolkata, India	1568	2 (0·13%)	2014	1 (0·05%)	2·6 (0·34–19·6)‡
All Sites Combined	9439	190 (2·0%)	13 129	37 (0·28%)	8·5 (5·8–12·5)†

OR=odds ratio. *Includes all enrolled patients with moderate-to-severe diarrhoea and their matched controls who met one of the following criteria: (1) the child was known to have died within 90 days of enrolment; (2) the child completed a follow-up visit between 50–90 days after enrolment; or (3) the child was located after day 90 and found to be alive. Note that the follow-up period in Kenya was actually 49–91 days. †ORs and 95% CIs from weighted conditional logistic regression; Mozambique (p<0·0001), The Gambia (p<0·0001), Kenya (p<0·0001), Mali (p=0·002), Pakistan (p=0·051), and p<0·0001 for all sites combined. ‡OR (unmatched) and 95% CI from a likelihood score method;[16] p=0·004 for Bangladesh and p=0·58 for India, two-sided Fisher's exact test.

Table 4: **Mortality in children with moderate-to-severe diarrhoea and their matched controls between enrolment and follow-up, by site**

with a higher risk of dying in patients with moderate-to-severe diarrhoea were ST-ETEC and typical EPEC in infants and *Cryptosporidium* in toddlers (table 5). Adjusting for site, enrolment HAZ was inversely associated with risk of dying in patients with moderate-to-severe diarrhoea in all age groups, as follows: 0–11 months HR 0·62 (95% CI 0·54–0·72, p<0·0001); 12–23 months HR 0·74 (95% CI 0·63–0·87, p=0·0002); and 24–59 months HR 0·47 (95% CI 0·38–0·57, p<0·0001).

Mean HAZ at enrolment in patients with moderate-to-severe diarrhoea and controls was considerably below the WHO reference for infants and, with one exception, deviated further from the reference at older ages (table 6). Linear growth faltering was especially marked at the Pakistan site. When we compared mean enrolment HAZ in cases of moderate-to-severe diarrhoea and their matched controls at the seven sites and three age strata, HAZ was significantly lower in cases than in controls in only two of 21 comparisons, both in infants (table 6). In pooled analysis, enrolment HAZ of cases and controls did not differ in the two older age strata but was significantly lower in infants with moderate-to-severe diarrhoea than in controls. Importantly, HAZ scores of moderate-to-severe diarrhoea cases decreased between enrolment and follow-up (ie, negative δHAZ), with only one exception (Malian children aged 24–59 months); the decline was significantly greater in patients with moderate-to-severe diarrhoea than in controls in most site-age strata and in all age strata in the pooled analysis, after adjusting for enrolment HAZ and time to follow-up (table 6).

Discussion

Using a comprehensive panel of microbiological assays, GEMS was performed to better define the incidence, aetiology, and clinical outcome of moderate-to-severe paediatric diarrhoea in seven sites, located in regions where more than 80% of deaths in children younger than 5 years occur,[17] and representing a range of health indicators (eg, malaria, HIV prevalence), health-care

	Pathogen present		Pathogen absent		Unadjusted†		Adjusted‡	
	Total	Number of deaths (%)	Total	Number of deaths (%)	HR (95% CI)	p value	HR (95% CI)	p value
0–11 months								
Typical EPEC	375	24 (6·4%)	3654	83 (2·3%)	2·8 (1·7–4·5)	<0·0001	2·6 (1·6–4·1)	0·0001
ST-ETEC	256	12 (4·7%)	3773	95 (2·5%)	2·0 (1·1–3·8)	0·03	1·9 (0·99–3·5)	0·05
E histolytica	115	5 (4·3%)	3914	102 (2·6%)	1·6 (0·64–4·2)	0·30	··	··
LT-ETEC	171	6 (3·5%)	3858	101 (2·6%)	1·2 (0·52–2·8)	0·67	··	··
Atypical EPEC	162	5 (3·1%)	3867	102 (2·6%)	1·0 (0·41–2·6)	0·93	··	··
Cryptosporidium	624	19 (3·0%)	3405	88 (2·6%)	1·2 (0·71–2·0)	0·52	··	··
EAEC	999	27 (2·7%)	3030	80 (2·6%)	0·92 (0·58–1·5)	0·71	··	··
Giardia	377	10 (2·7%)	3652	97 (2·7%)	1·1 (0·57–2·2)	0·74	··	··
Rotavirus	1016	25 (2·5%)	3013	82 (2·7%)	0·81 (0·51–1·3)	0·37	··	··
Aeromonas	251	5 (2·0%)	3778	102 (2·7%)	0·59 (0·24–1·5)	0·25	··	··
C jejuni	454	7 (1·5%)	3575	100 (2·8%)	0·55 (0·25–1·2)	0·13	··	··
12–23 months								
E histolytica	100	5 (5·0%)	3104	55 (1·8%)	3·5 (1·3–9·2)	0·01	2·1 (0·76–6·0)	0·15
Cryptosporidium	374	15 (4·0%)	2830	45 (1·6%)	2·6 (1·4–4·8)	0·002	2·3 (1·3–4·3)	0·006
EAEC	578	18 (3·1%)	2627	42 (1·6%)	1·7 (0·96–3·0)	0·07	··	··
Typical EPEC	232	7 (3·0%)	2973	53 (1·8%)	2·0 (0·87–4·5)	0·11	··	··
ST-ETEC	249	6 (2·4%)	2956	54 (1·8%)	1·2 (0·48–3·1)	0·68	··	··
Shigella	485	8 (1·6%)	2720	52 (1·9%)	0·89 (0·42–1·9)	0·77	··	··
Giardia	693	7 (1·0%)	2511	53 (2·1%)	0·51 (0·22–1·2)	0·12	··	··

HR=hazard ratio. ST=heat stable toxin. LT=heat labile toxin. ST-ETEC=either ST or ST/LT producing enterotoxigenic *E coli*. LT-ETEC=ETEC producing only LT. EPEC=enteropathogenic *E coli*. EAEC=enteroaggregative *E coli*. *Includes all enrolled patients with MSD and their matched controls who met one of the following criteria: (1) the child was known to have died within 90 days of enrolment. (2) the child completed a follow-up visit between 50–90 days after enrolment; or (3) the child was located after day 90 and found to be alive. Note that the follow-up period in Kenya was actually 49–91 days. †From Cox regression on data pooled from all sites, with presence or absence of pathogen as the only covariate; only pathogens that were isolated from at least five patients with MSD who died were included. ‡Adjusted for other pathogens, as well as study site; from Cox regression on data pooled from all sites, and including as covariates pathogens with p <0·05 in unadjusted analysis and dichotomous variables for site.

Table 5: **Weighted unadjusted and adjusted hazard ratios for selected pathogens and risk of death between enrolment and follow-up,in cases of moderate-to-severe diarrhoea (MSD)***

accessibility, economic development, and environmental conditions. By including matched control children without diarrhoea, we derived burden estimates adjusted for the occurrence of asymptomatic colonisation with enteropathogens often seen in children living in faecally contaminated environments,[18] and derived an AF for every pathogen that was independently associated with moderate-to-severe diarrhoea in regression models, adjusting for interactions and confounding effects of co-infecting enteropathogens.[6] Pathogen-specific adjusted AFs estimate the proportion of moderate-to-severe diarrhoea at our sites that could be prevented with targeted interventions such as effective vaccines.

Moderate-to-severe diarrhoea was common in the paediatric populations studied, producing more than 20 episodes per 100 child-years during each of the first 2 years of life. Three findings concerning children with moderate-to-severe diarrhoea are noteworthy. First, despite the wide array of putative pathogens that we detected, a small number contributed most attributable moderate-to-severe diarrhoea cases: rotavirus, *Cryptosporidium*, ST-ETEC, and *Shigella*, and, to a lesser extent, adenovirus 40/41. Several other pathogens were important only in Asia (*Aeromonas*) or Asia and Mozambique (*V cholerae* O1, *C jejuni*). Our findings support the notion that in Asia, *Aeromonas* causes diarrhoeal disease in young children.[19] Second, children with moderate-to-severe diarrhoea experienced a substantial nutritional insult, evidenced by significantly more linear growth faltering during the follow-up period compared with their matched controls, even though, in 19 of the 21 site-strata, the mean enrolment HAZ scores of cases and controls were similar. Finally, compared with controls, moderate-to-severe diarrhoea cases had greatly (8·5 times) increased risk of dying during the follow-up period, and the risk was inversely associated with enrolment HAZ. Although risk of dying in patients with diarrhoea was greatest in GEMS sites with high HIV prevalence (Mozambique, Kenya), mortality in patients with moderate-to-severe diarrhoea was also substantial in our low HIV-prevalence rural sub-Saharan site (The Gambia), and moderate-to-severe diarrhoea was significantly associated with increased risk of death at all sites except India. Importantly, most deaths occurred outside health facilities and were detected only because the study

	0–11 months	12–23 months	24–59 months
Basse, The Gambia	293 cases; 423 controls	322 cases; 445 controls	127 cases; 237 controls
Enrolment HAZ			
Cases	−0·81 (−0·97 to −0·66)	−1·31 (−1·46 to −1·16)	−1·77 (−1·99 to −1·56)
Controls	−0·63 (−0·81 to −0·44)	−1·32 (−1·49 to −1·16)	−1·49 (−1·72 to −1·26)
δHAZ			
Cases	−0·31 (−0·38 to −0·23)	−0·29 (−0·34 to −0·23)‖	−0·07 (−0·14 to −0·01)¶
Controls	−0·34 (−0·42 to −0·26)	−0·14 (−0·19 to −0·08)‖	0·03 (−0·04 to 0·09)¶
Bamako, Mali	521 cases; 521 controls	520 cases; 528 controls	501 cases; 512 controls
Enrolment HAZ			
Cases	−0·60 (−0·72 to −0·48)*	−1·07 (−1·19 to −0·95)	−1·14 (−1·25 to −1·03)
Controls	−0·38 (−0·52 to −0·25)*	−1·04 (−1·15 to −0·92)	−1·08 (−1·18 to −0·97)
δHAZ			
Cases	−0·28 (−0·33 to −0·22)‡	−0·07 (−0·10 to −0·04)	0·07 (0·05 to 0·09)
Controls	−0·24 (−0·29 to −0·19)‡	−0·04 (−0·07 to −0·004)	0·09 (0·07 to 0·10)
Manhiça, Mozambique	230 cases; 405 controls	112 cases; 199 controls	59 cases; 104 controls
Enrolment HAZ			
Cases	−1·24 (−1·42 to −1·07)*	−1·65 (−1·90 to −1·39)	−1·55 (−1·93 to −1·18)
Controls	−1·03 (−1·17 to −0·90)*	−1·58 (−1·83 to −1·33)	−1·74 (−2.03 to −1·44)
δHAZ			
Cases	−0·21 (−0·31 to −0·11)‡	−0·41 (−0·50 to −0·31)‖	−0·15 (−0·24 to −0·06)‡
Controls	−0·12 (−0·20 to −0·05)‡	−0·07 (−0·15 to 0·01)‖	−0·003 (−0·06 to 0·05)‡
Nyanza Province, Kenya	560 cases; 560 controls	340 cases; 523 controls	339 cases; 519 controls
Enrolment HAZ			
Cases	−0·96 (−1·06 to −0·86)	−1·54 (−1·68 to −1·40)	−1·65 (−1·78 to −1·52)*
Controls	−0·93 (−1·03 to −0·83)	−1·64 (−1·78 to −1·50)	−1·76 (−1·87 to −1·66)*
δHAZ			
Cases	−0·35 (−0·40 to −0·30)‖	−0·30 (−0·34 to −0·26)‖	−0·13 (−0·16 to −0·11)‖
Controls	−0·21 (−0·25 to −0·17)‖	−0·10 (−0·14 to −0·06)‖	−0·008 (−0·03 to 0·01)‖
Kolkata, India	623 cases; 635 controls	542 cases; 551 controls	295 cases; 690 controls
Enrolment HAZ			
Cases	−1·02 (−1·10 to −0·93)	−1·49 (−1·60 to −1·39)	−1·68 (−1·83 to −1·54)
Controls	−1·02 (−1·11 to −0·93)	−1·41 (−1·51 to −1·31)	−1·70 (−1·81 to −1·59)
δHAZ			
Cases	−0·29 (−0·32 to −0·25)‡	−0·14 (−0·17 to −0·11)	−0·05 (−0·08 to −0·02)
Controls	−0·34 (−0·38 to −0·29)‡	−0·11 (−0·14 to −0·09)	−0·02 (−0·04 to 0·005)
Mirzapur, Bangladesh	523 cases; 824 controls	455 cases; 712 controls	352 cases; 780 controls
Enrolment HAZ			
Cases	−1·06 (−1·16 to −0·96)	−1·32 (−1·43 to −1·22)	−1·50 (−1·61 to −1·39)
Controls	−1·07 (−1·15 to −0 · 99)	−1·36 (−1·45 to −1·28)	−1·53 (−1·63 to −1·43)
δHAZ			
Cases	−0·28 (−0·32 to −0·23)¶	−0·18 (−0·21 to −0·14)‖	−0·15 (−0·18 to −0·12)‖
Controls	−0·16 (−0·20 to −0·12)¶	−0·06 (−0·09 to −0·03)‖	−0·02 (−0·03 to 0·0001)‖
Karachi (Bin Qasim Town), Pakistan	397 cases; 397 controls	291 cases; 453 controls	170 cases; 353 controls
Enrolment HAZ			
Cases	−1·61 (−1·74 to −1·48)	−2·16 (−2·34 to −1·99)	−2·43 (−2·64 to −2·22)
Controls	−1·50 (−1·63 to −1·37)	−2·06 (−2·21 to −1·92)	−2·28 (−2·44 to −2·13)

(Continues on next page)

included a follow-up home visit about 60 days after enrolment.[3,7] Although we did not undertake surveillance to identify interim events affecting the child's outcome, our findings suggest that moderate-to-severe diarrhoea might contribute to the causal pathway to death, probably influenced by its associated nutritional derangement.

Evidence from GEMS that rotavirus is the most common cause of moderate-to-severe diarrhoea at every site during the first year of life renders optimism that the decision by the GAVI Alliance to make rotavirus vaccines available to the world's poorest countries[20] will substantially benefit global child health. Surveillance at GEMS sites could ascertain the effect of programmatic use of rotavirus vaccine including whether vaccine-derived protection persists beyond infancy. Although the point estimate of rotavirus vaccine efficacy recorded in field trials is distinctly lower in resource-poor countries than in resource-rich settings,[21] high levels of rotavirus vaccine coverage might nevertheless achieve profound effects.[22]

Remarkably, *Cryptosporidium* was a significant pathogen at all sites regardless of HIV prevalence, and the second most common pathogen in infants. Although often reported as a cause of life-threatening illness in individuals with HIV/AIDS and of diarrhoea and malnutrition in young children in sub-Saharan Africa,[23] the disease burden in the general paediatric population has been poorly quantified (particularly in Asia). The importance of *Cryptosporidium* in GEMS, and its association with death during the ensuing 2–3 months in toddlers aged 12–23 months, is consistent with previous findings of a cohort study in Guinea-Bissau,[24] a low HIV-prevalent area. These findings highlight the need to develop resources to diagnose, treat, and prevent cryptosporidiosis in resource-poor settings.

EPEC has been associated with acute and persistent diarrhoea leading to nutritional faltering and death.[25] However, the frequent detection in children without diarrhoea and high prevalence of breastfeeding (known to be protective) in many study populations are among the factors that have obfuscated estimates of EPEC disease burden in developing countries.[26] The ability of typical EPEC to cause diarrhoea is well established;[27] the virulence of atypical EPEC is less certain.[28] In GEMS we noted no association between atypical EPEC and moderate-to-severe diarrhoea, whereas typical EPEC was significantly associated with moderate-to-severe diarrhoea during the first 2 years of life at one site (Kenya). When we limited our analysis to cases, typical EPEC was significantly associated with death in infants aged 0–11 months. Efforts to prevent diarrhoea-associated morbidity and mortality might have to include pathogens such as typical EPEC that are not strongly associated with moderate-to-severe diarrhoea overall, but, when present in patients with moderate-to-severe diarrhoea, seem to be disproportionately associated with poor outcomes. Since linear growth faltering was a risk factor

for death in GEMS, nutritional rehabilitation should be part of case management algorithms for diarrhoea.

Limitations of the GEMS design have been described,[4] and others are noted here. For one, antibiotic use before the SHC visit could have diminished the yield of bacterial cultures. Conversely, pathogens identified with molecular tests might be overdiagnosed in cases and controls.[29] The GEMS sites are endemic for many enteropathogens, creating an environment in which the same enteropathogens are commonly detected in controls without diarrhoea, which underestimates AF.[18,30] That children with moderate-to-severe diarrhoea are at high risk for linear growth faltering and death is of utmost importance and demands further investigation. Nonetheless, one must be circumspect in assuming causality, particularly because our study did not define the contribution of intercurrent illnesses, nor did it factor in comorbidities (eg, HIV, malnutrition) that might have contributed both to the episode of moderate-to-severe diarrhoea and the adverse outcome. Our matched-pair analysis of the effect of an episode of moderate-to-severe diarrhoea on linear growth required exclusion of about 20% of individuals (appendix), which could theoretically introduce bias. Therefore, it was reassuring that baseline HAZ scores did not differ significantly between included and excluded children, nor did socio-demographic variables (appendix).

Finally, since the use of SHCs by children in the DSS population was low, our estimates of overall and pathogen-specific moderate-to-severe diarrhoea incidence might be imprecise. However, except for India, the site-specific incidence rates of rotavirus moderate-to-severe diarrhoea for infants (1·9 to 10·0 episodes per 100 child-years) were similar to published estimates of moderate-to-severe rotavirus diarrhoea (albeit derived using various definitions) from developing countries in Africa and south Asia.[31,32] The India site had the lowest SHC use for moderate-to-severe diarrhoea (r value), possibly due to some unknown source of bias, and a lower r increases the variability of the point estimate. Second, in the crowded slums of Kolkata, where there are open sewage drains and the highest population density of any GEMS site (18 601 individuals per km^2),[33] spread of enteric pathogens such as rotavirus and *Shigella* might be enhanced through several methods of transmission.[34] Finally, despite extensive training and standardisation of clinical criteria for moderate-to-severe diarrhoea, some site-to-site variation occurred. In India, 50% of DSS children with diarrhoea seeking care at the SHC were deemed to have moderate-to-severe diarrhoea, more often than any other site except Kenya. These factors might have contributed to higher estimates of both moderate-to-severe diarrhoea incidence and the proportion of moderate-to-severe diarrhoea attributable to specific pathogens in India than at other GEMS sites.

Our results documenting the substantial burden of moderate-to-severe diarrhoea in sub-Saharan Africa and south Asia and its close association to malnutrition show that preventive strategies targeting as few as four pathogens could potentially reduce this disease and its sequelae by about 40% during the first 2 years of life. Accordingly, an urgent need exists to accelerate introduction of existing interventions with proven effectiveness, such as rotavirus

	0–11 months	12–23 months	24–59 months
(Continued from previous page)			
δHAZ			
Cases	–0·26 (–0·32 to –0·20)	–0·19 (–0·23 to –0·14)‖	–0·05 (–0·09 to –0·006)¶
Controls	–0·25 (–0·31 to –0·19)	–0·15 (–0·19 to –0·11)‖	0·03 (–0·001 to 0·05)¶
All sites combined	3147 cases; 3765 controls	2582 cases; 3411 controls	1843 cases; 3195 controls
Enrolment HAZ			
Cases	–0·98 (–1·03 to –0·93)†	–1·43 (–1·48 to –1·37)	–1·60 (–1·66 to –1·53)
Controls	–0·86 (–0·91 to –0·81)†	–1·41 (–1·47 to –1·35)	–1·57 (–1·63 to –1·51)
δHAZ			
Cases	–0·29 (–0·31 to –0·27)‖	–0·19 (–0·21 to –0·18)‖	–0·06 (–0·07 to –0·04)‖
Controls	–0·25 (–0·27 to –0·23)‖	–0·09 (–0·11 to –0·08)‖	0·02 (0·009 to 0·03)‖

Data are weighted mean (95% CI). δHAZ denotes HAZ at the follow-up visit minus HAZ at enrolment. *The follow-up period in Kenya was 49–91 days. Enrolment HAZ in cases *vs* controls was compared by weighted paired *t* test: *p=0·01–0·04 and †p=0·0009; all other p values >0·05. ΔHAZ in cases *vs* controls was compared by weighted linear regression, adjusting for enrolment HAZ and duration to follow-up: ‡p=0·01 to <0·05; ¶p=0·0001–0·008; ‖p<0·0001, all other p values >0·05.

Table 6: **Comparison of enrolment length or height for age *Z* score (HAZ), and change in HAZ (ΔHAZ) between enrolment and follow-up 50–90 days later,* between cases with moderate-to-severe diarrhoea and their matched controls, by site**

Panel: Research in context

Systematic review

We searched PubMed for English, French, Spanish, and Portuguese publications using various combinations of the terms "diarrhea," "gastroenteritis," "diarrheal disease," "pediatric", "etiology", "microbiology", "growth faltering", and "malnutrition". We focused mainly on studies published since 1980 but included older reports where relevant. To identify additional publications we perused the reference lists of the original and review articles. Epidemiological studies were critically reviewed to detect methodological limitations and microbiological techniques were scrutinised. We also made judgments about the interest and relevance of studies for the well informed general clinician and public health practitioner.

From 1980 through roughly 2004, various case-control and small cohort studies investigated the aetiology of paediatric diarrhoea in low-income countries. Many studies had methodological inadequacies and arrived at disparate conclusions, making it difficult to prioritise the relative importance of different pathogens. Thus, there was no consensus on what specific diarrhoeal disease pathogens should be targeted for prevention. By contrast, agreement existed on the need for a well designed study to obtain information on the aetiology and burden of more severe forms of diarrhoeal disease to guide global investment and implementation decisions.

Interpretation

The Global Enteric Multicenter Study (GEMS) was designed to overcome drawbacks of earlier studies and determine the aetiology and population-based burden of paediatric diarrhoeal disease. Our findings demonstrate that interventions targeting only five pathogens can substantially reduce the burden of moderate-to-severe diarrhoea. GEMS data will guide investment and help prioritise strategies to mitigate the morbidity and mortality of paediatric diarrhoeal disease.

vaccination and adjunct treatment of acute diarrhoea with zinc, to develop additional strategies with potential high impact, such as new vaccines, and to revitalise diarrhoeal disease case management algorithms shown to reduce malnutrition (panel).[35,36]

Contributors

MML conceived the project and acquired the grant funds. MML, KLK, and JPN designed the protocol. WCB did the statistical analysis with YW, DN, HS, THF, and KM. KLK, JPN, DN, THF, SP, LB, SOS, DSu, RFB, ASGF, AKMZ, RAA, DSah, PLA, EDM, and CEOR planned and supervised the study. DSan, SK, RO, SKD, SA, FQ, AA, TN, and SA coordinated clinical data collection; BT, TR, JBO, JOO, AH, SQ, MA, IM, and RMR-B did the laboratory assays; and UO, BM, MJH, and KB participated in data management. KLK, WCB, DN, THF, YW, KM, JPN, and MML had full access to all the data in the study and did data analysis; KLK wrote the report with input from all authors and had final responsibility for the decision to submit for publication. All authors reviewed the draft and approved the decision to submit for publication.

Conflicts of interest

We declare that we have no conflicts of interest.

Acknowledgments

This study was funded by the Bill & Melinda Gates Foundation. We thank the families who participated in these studies, the project field staff for their hard work and dedication, and to the physicians and administration at every site who generously provided facilities for the conduct of the study.

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